

EX.NO: 13**IMPLEMENTATION OF VARIOUS CHARTS****AIM:**

To implement different types of charts in R, including bar charts, pie charts, and line charts, using built-in R functions.

PROCEDURE:

1. Install and Load Required Packages – Ensure R's graphical libraries (like ggplot2) are installed if needed.
2. Create Sample Data – Define datasets for visualization.
3. Generate Different Charts – Use functions like barplot(), pie(), and plot() to create bar, pie, and line charts.
4. Display Output – Execute the script and visualize the charts.

SOURCE CODE:

```
# Bar Chart
```

```
data <- c(20, 30, 50, 40, 60)
```

```
names <- c("A", "B", "C", "D", "E")
```

```
barplot(data, names.arg=names, col="blue", main="Bar Chart Example", xlab="Categories",  
ylab="Values")
```

```
# Pie Chart
```

```
values <- c(25, 35, 40)
```

```
labels <- c("Apple", "Banana", "Cherry")
```

```
pie(values, labels=labels, col=rainbow(length(values)), main="Pie Chart Example")
```

```
# Line Chart
```

```
x <- 1:10
```

```
y <- x^2
```

```
plot(x, y, type="o", col="red", main="Line Chart Example", xlab="X-axis", ylab="Y-axis")
```

OUTPUT:

1. Bar Chart – Displays a bar graph with labeled categories and corresponding values.
2. Pie Chart – Shows a pie chart with colored segments for different items.
3. Line Chart – Plots a line graph representing the relation between x and y values.

Result:

The implementation of different charts in R was successfully executed, and the required visualizations were generated.